U.S. Software

Raimo's Trend Spotter – 31st Edition of Tracking Web Traffic and PyPi Data in Software

Q3 web traffic growth softened across much of software. Infrastructure growth generally decelerated on a q/q basis. In data platforms, Snowflake weakened. E-commerce visits saw strong growth. In email marketing, Klaviyo continued to exceed Mailchimp in visits.

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Executive Summary

As software companies continue to move to a cloud delivery model, web login pages will continue to be prominent for many of our covered companies. Therefore, using login URLs, we can track web traffic to those pages to gauge overall popularity and usage trends for cloud products.

In the 31st edition of our quarterly Raimo's Trend Spotter, we share trends and insights we have identified in the software world using web traffic data points, namely total visits and unique visits.

What's incremental in this edition?

1. Cloud infrastructure growth generally decelerated. Azure's total visits were flat q/q (vs. +0.4% last quarter), AWS went down 2.6% q/q (vs. +1.2% last quarter), and GCP went down 7.7% q/q (vs. -1.5% last quarter). DOCN also fell by 3.6% q/q (vs. +3.7% last quarter).
2. Observability results were mixed. Dynatrace's web traffic declined 0.8% q/q (vs. +17.8% last quarter) but remained up +45.8% y/y, Splunk declined 19.6% q/q (vs. +7.3% last quarter), and Elastic was down 4.3% q/q (vs. -4.1% last quarter).
3. For data platforms, Snowflake remained the market leader, although total visits were down 28% q/q (vs. +8% last quarter). Other vendors showed mixed trends, with Databricks accelerating to +17% q/q (vs. 12% last quarter) and Teradata declining 1.4% q/q (vs. -1.7% last quarter).
4. For NoSQL database sequential trends, MongoDB Atlas traffic turned negative for both total and unique visits on a y/y basis, with total visits declining 19% q/q and -17% y/y (vs. +3%

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and +7% last quarter) and unique visits declining 17% q/q and -4% y/y (vs. +4% and +22% last quarter).

5. ChatGPT total visits increased this quarter, a reversal of the continuous decline in Q1 and Q2, and we note an upward momentum near the end of Q3, although the number of unique visitors declined in this quarter.

6. In e-commerce, Shopify remained the leader in traffic volume (~10x the next company we track). Total web traffic for e-commerce increased q/q (+10.3% vs. -1.0% last quarter) and accelerated on a y/y basis (+22.4% vs. +13.6% last quarter).

7. For email marketing, Klaviyo continued to see the highest total volume in Q3, after surpassing Mailchimp (owned by Intuit) earlier this year, although both declined on a q/q basis. Total visits for Mailchimp were down 2.7% q/q (vs. -11.3% last quarter), compared to Klaviyo being down 1.2% q/q (vs. -1.8% last quarter).

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Web Traffic Data – Overview

In this section, we use web traffic data to uncover different trends in the software space. As many of these products are now being delivered over the web, we can track the traffic data that goes to specific URLs, which enables us to gauge the popularity and usage trends in software.

Below, we look at the following web traffic trends:

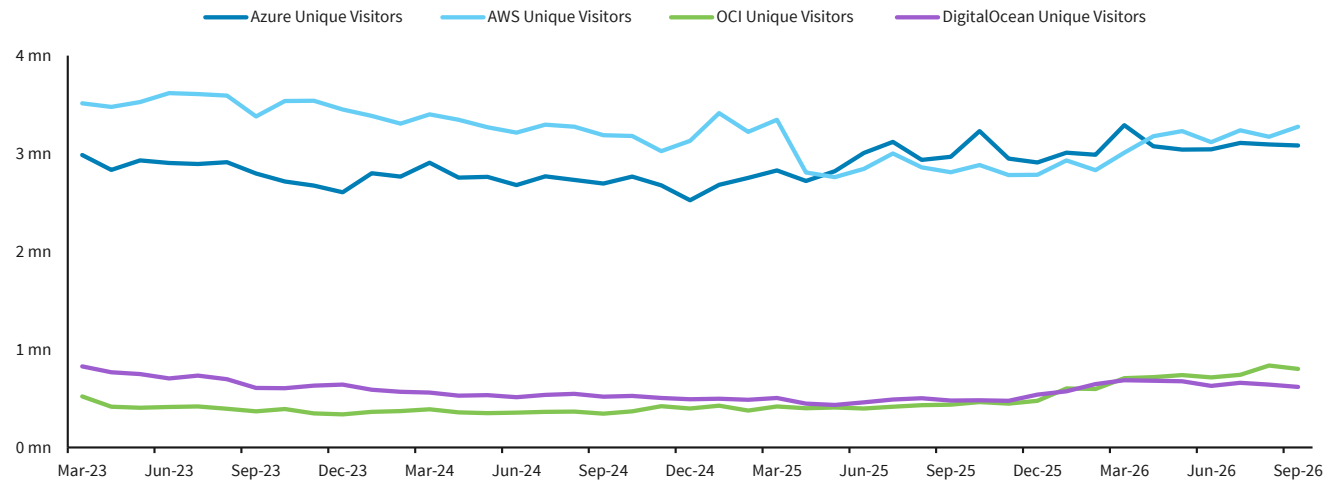
- Cloud IaaS Vendor Console Traffic;
- AI Update - ChatGPT and Bing Visitors;
- NoSQL Databases - MongoDB Atlas Traffic;
- Cloud Data Warehouse/Platform Traffic;
- Observability Vendors - Elastic, Splunk, Dynatrace, and Datadog Traffic;
- E-Commerce Platform Traffic;
- Email Marketing Traffic.

Cloud Infrastructure Trends

The move to the cloud remains one of the most powerful trends in software. Hence, we look at the cloud infrastructure main landing pages (console page) to track the trends in popularity that each cloud is seeing in the market.

- Azure unique visits were up +1.4 q/q vs. down -1% last quarter. On a y/y basis, unique visits were up +3%, compared to up +7% last quarter.
- AWS unique visits were up +2% q/q vs. up +9% q/q last quarter, and on a y/y basis, growth was up +12%, compared to the quarter prior at up +13%.
- Google unique visits numbers were down -3.3% q/q vs. flat last quarter. On a y/y basis, growth was +10% vs. +16% last quarter.
- OCI was up +10% q/q, compared to a +14% increase the prior quarter. On a y/y basis, OCI saw a +85% increase compared to a +80% increase last quarter.
- DigitalOcean saw growth fell to -3% growth q/q vs. an increase of +4% last quarter. On a y/y basis, visit growth was up +31% vs. +48% last quarter.

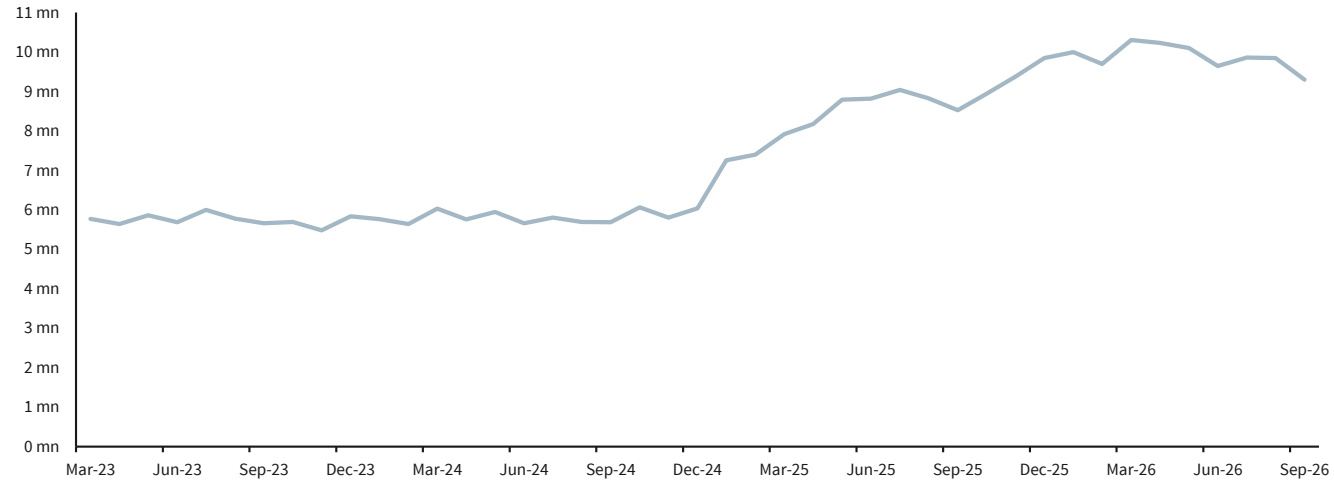
FIGURE 1. Cloud Infrastructure - Monthly Unique Visits



Source: Barclays Research, Similarweb

We track Google Cloud separately (see figure below) as the scale of unique visitors is larger compared to the rest of the group. We note that we use console.cloud.google.com as our reference point to track web traffic for Google Cloud. However, we note that this URL represents users that have authenticated to get onto the Google Cloud platform, and also represents users that are leveraging BigQuery and Firebase, opening compute, logs, etc., and thus the numbers are higher compared to leaders like Azure and AWS.

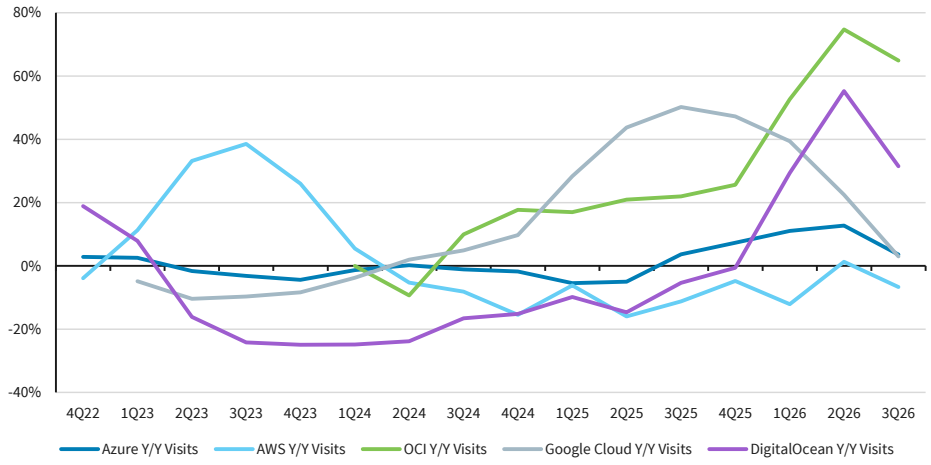
FIGURE 2. Google Cloud - Monthly Unique Visits



Source: Barclays Research, Similarweb

We also looked at the y/y growth changes in total visits for the cloud infrastructure names. In Figure 3, we can see OCI (+65% y/y vs. +75% in 2Q26) and DigitalOcean (+32% y/y vs. 5% in 2Q26) growth continued to lead the way, although both decelerated. GCP was +3% y/y vs. 23% last quarter, Azure was + 4% y/y vs. 12% last quarter, and AWS fell by -7% y/y vs. +1% last quarter. We note that OCI's growth is due to the smaller base compared to the rest of the group.

FIGURE 3. Cloud Infrastructure - Total Visits Y/Y Change



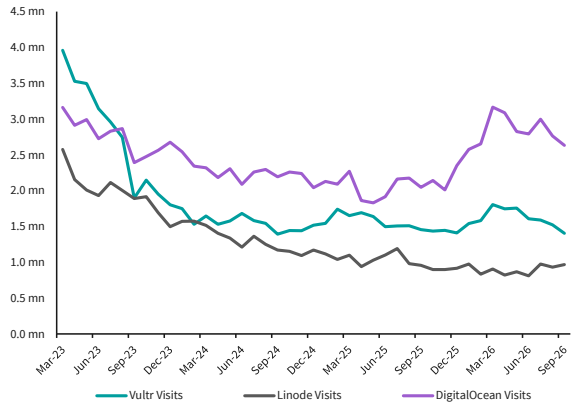
Source: Barclays Research, Similarweb

Earlier this year, we also started to track Linode (owned by Akamai; public, not covered) and Vultr (private, not covered), to compare growth vs. DigitalOcean as all play more in the lower end of the market (see graphs below).

For total visits, DigitalOcean was down -4% q/q (vs. +4% in 2Q26) and was up +32% y/y (vs. +55% in 2Q26); Vultr was down -12% q/q (vs. +4% in 2Q26) and was up +1% y/y (vs. +6% in 2Q26); Linode was up +15% q/q (vs. -8% in 2Q26) and was down -8% y/y (vs. -19% in 2Q26).

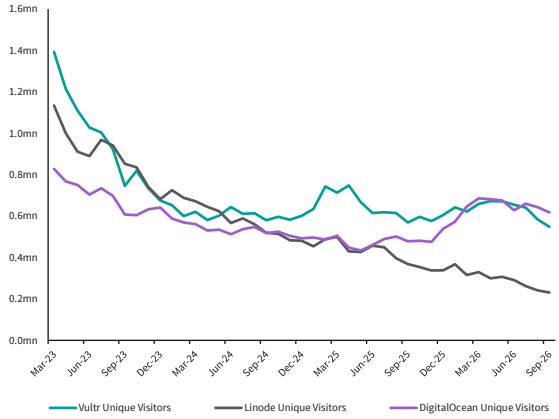
For unique visits, DigitalOcean was down -3% q/q (vs. +4% in 2Q26) and was up 31% y/y (vs. +48% in 2Q26); Vultr was down -11% q/q (vs. +4% in 2Q26) and was down -2% y/y (vs. -2% in 2Q26); Linode was down -18% q/q (vs. -11% in 2Q26) and was down -39% y/y (vs. -32% in 2Q26).

FIGURE 4. Cloud Service Provider - Monthly Total Visits



Source: Barclays Research, Similarweb

FIGURE 5. Cloud Service Provider - Monthly Unique Visits



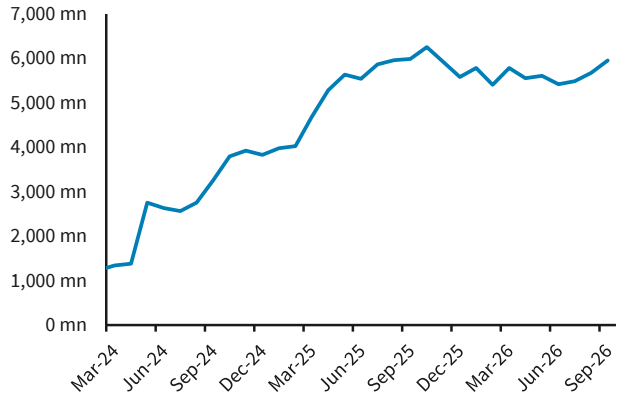
Source: Barclays Research, Similarweb

AI Update – ChatGPT and Bing Visitors

ChatGPT

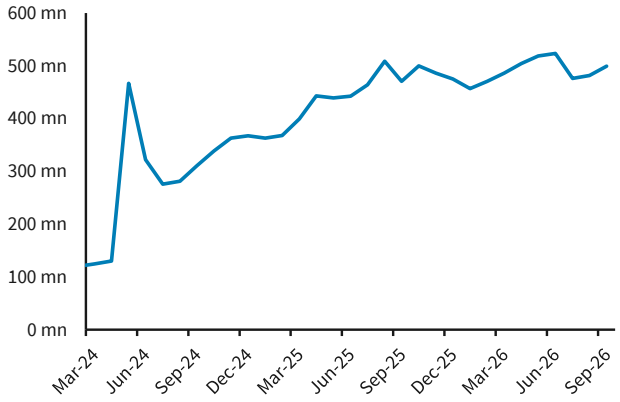
We continue to track total and unique visits to ChatGPT. This quarter, we saw an increase in total visits and an upward momentum near the end of Q3, although a decrease in the number of unique visitors. We also note that this data does not take into account the ChatGPT iOS app, and as such may not be a fully comprehensive indicator of popularity increasing/decreasing.

FIGURE 6. ChatGPT Total Visits Per Month



Source: Barclays Research, Similarweb

FIGURE 7. ChatGPT Unique Visits Per Month

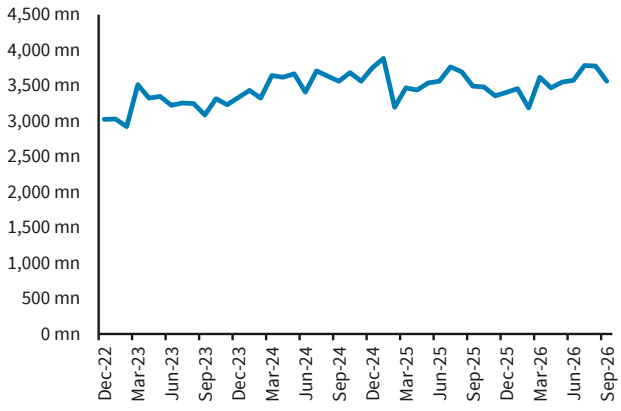


Source: Barclays Research, Similarweb

Bing

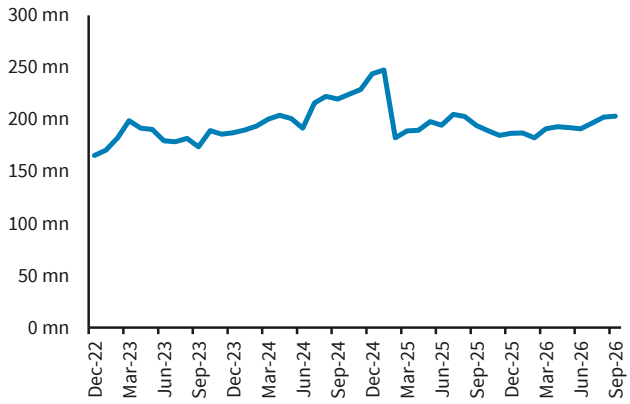
Given the integration of ChatGPT into Bing (owned by MSFT), we track total and unique visits to the Bing platform. Bing usage slightly increased this quarter, although still remaining far behind Google in terms of absolute usage numbers. Usage reached its peak in July 2026 with 3.79bn visits, and slightly declined near the end of Q3 to 3.56bn in September. Sequentially, total visits grew +5% q/q vs. +3% last quarter. On a y/y basis, total visits and unique visits changed +2% and remained flat, respectively.

FIGURE 8. Bing Total Visits



Source: Barclays Research, Similarweb

FIGURE 9. Bing Unique Visits



Source: Barclays Research, Similarweb

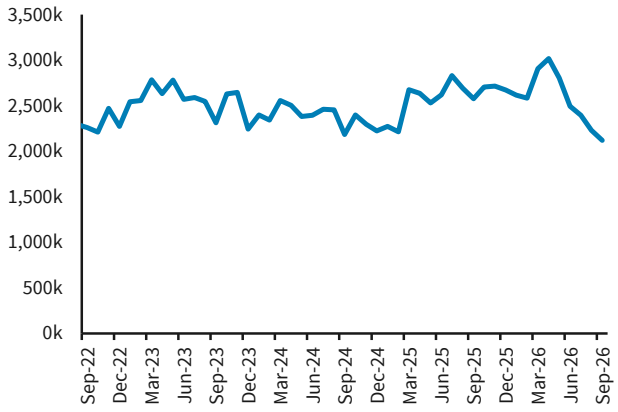
NoSQL Databases – MongoDB Atlas

MongoDB Atlas

A key driver of the MongoDB growth story is its database-as-a-service product, MongoDB Atlas. Hence, we track the web traffic (both total visits and unique visits) to the Atlas login page.

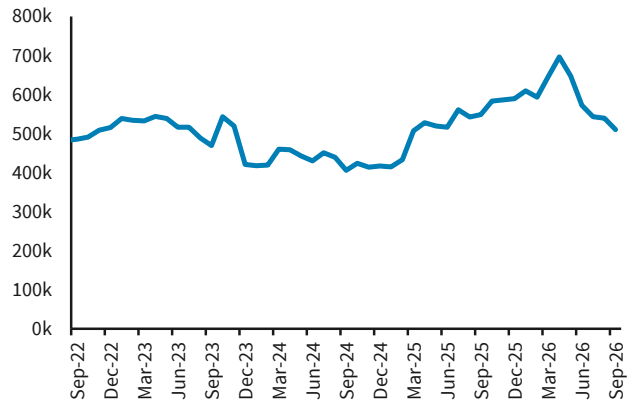
- Atlas traffic showed negative growth for both total and unique visits on a y/y basis this quarter. Visits continued to decline near the end of Q3 after peaking in this April.
- Total visits fell -19% q/q (vs. +3% last quarter) and were -17% y/y (vs. +7% last quarter). Quarterly unique visits were down -17% q/q (vs. +4% last quarter) and were down -4% y/y (vs. +22% last quarter).

FIGURE 10. MongoDB Atlas Login Page - Total Page Visits



Source: Barclays Research, Similarweb

FIGURE 11. MongoDB Atlas Login Page - Monthly Unique Visits



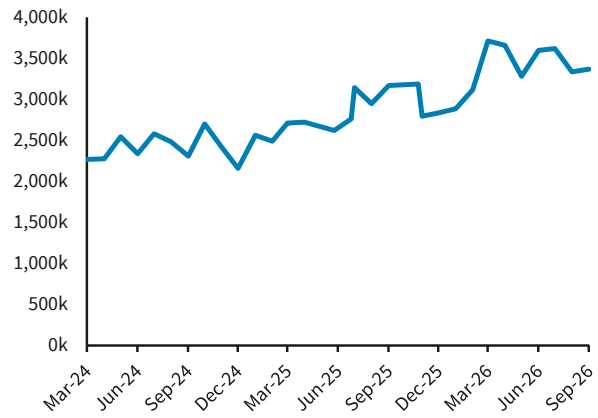
Source: Barclays Research, Similarweb

Cloud Data Warehouses/Platforms

Cloud data warehouses are becoming mainstream in the analytics software market. Hence, we track the traffic of Snowflake, Teradata, and Databricks (private, not covered) cloud login pages.

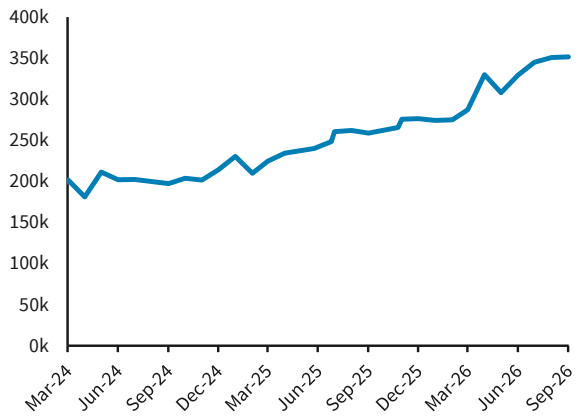
- Snowflake remains the clear market leader. We continue to break out Snowflake from the rest of the data to get a better picture of the underlying trends in its usage. Total visits were down -2% q/q (vs. +8% last quarter) and +12% y/y (vs. +30% last quarter), and unique visits were up +8% q/q (vs. +16% last quarter) and +34% y/y (vs. +34% last quarter).

FIGURE 12. Snowflake Monthly Total Visits



Source: Barclays Research, Similarweb

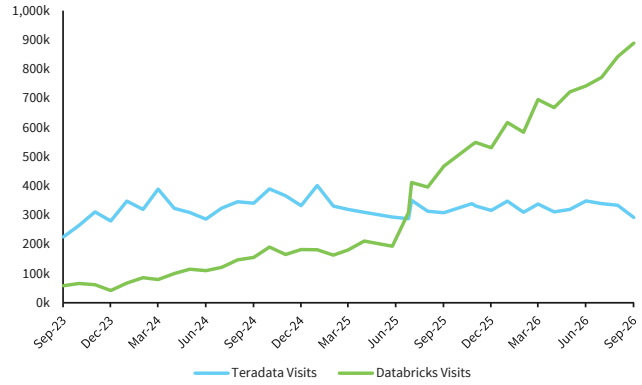
FIGURE 13. Snowflake Monthly Unique Visits



Source: Barclays Research, Similarweb

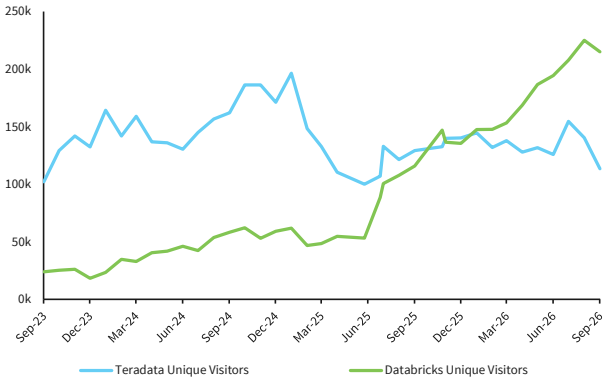
- We then looked at the other EDWs/data platforms we monitor. Databricks continued to gain the highest traction in terms of the number of total visits and monthly unique visits. Databricks total visits were up +17% q/q (vs. 12% last quarter), while unique visits were up +18% q/q (vs. +23% last quarter) and up +100% y/y (vs. +180% last quarter). Teradata saw another slight decline in total visits (-1.4% q/q vs. -1.7% last quarter), although an uptick in unique visitors (+6% q/q vs. -7% last quarter).

FIGURE 14. Cloud EDWs - Total Visits



Source: Barclays Research, Similarweb

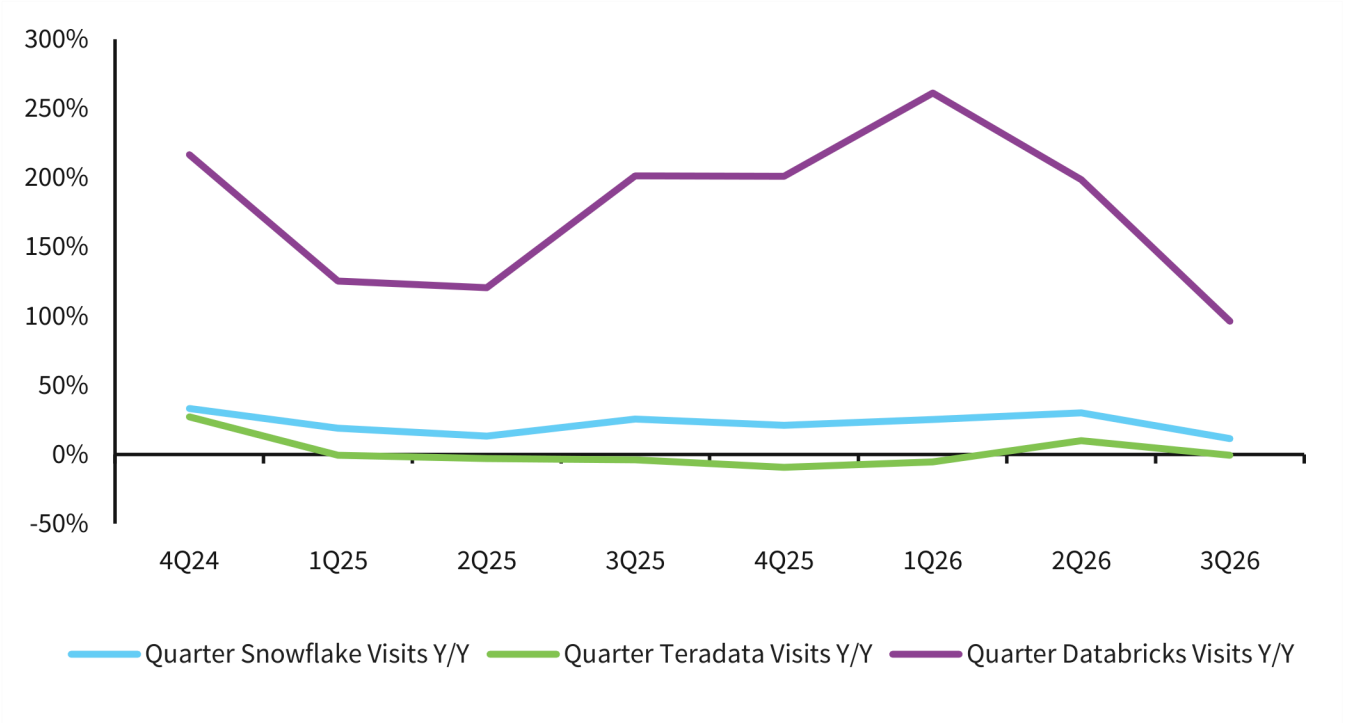
FIGURE 15. Cloud EDWs - Monthly Unique Visits



Source: Barclays Research, Similarweb

- We also looked at the y/y growth changes in total visits for the cloud data warehouse names in the figure below. Databricks continued to grow the most, although with a decelerated growth rate (+96% y/y vs. 199% y/y in 2Q26). Teradata saw a slight decline in y/y growth this quarter (-0.7% vs. +10.0% in 2Q26).

FIGURE 16. CDW Total Visits Y/Y Change



Does not include SNOW
Source: Barclays Research, Similarweb

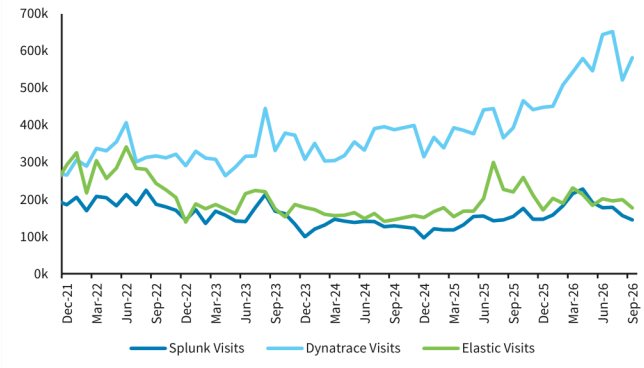
Observability Vendors

Elastic, Splunk, and Dynatrace Cloud Offerings

Elastic offers its technology as a fully-managed service named Elastic Cloud. Splunk and Dynatrace both offer cloud native software as part of their respective Observability suites.

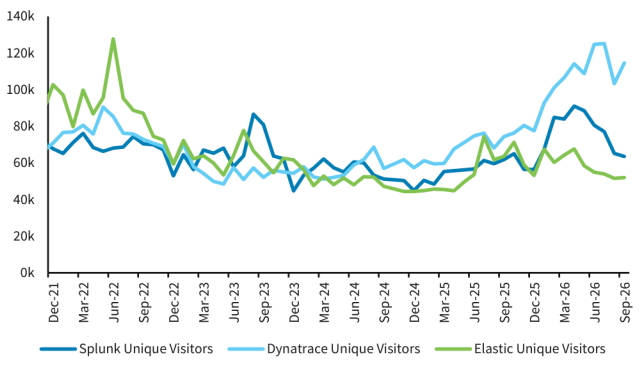
- For total visits, Elastic was down -4.3% q/q (vs. -4.1% last quarter), Splunk was down -19.6% q/q (vs. +7.3% last quarter), and Dynatrace was -0.8% q/q (vs. +17.8% last quarter). On a y/y basis, Elastic was -23.4% y/y (vs. +10.9% last quarter), Splunk was +8.7% y/y (vs. +35.4% last quarter), and Dynatrace was +45.8% y/y (vs. +47.0% last quarter).
- For unique visits, Elastic was down -12.9% q/q (vs. -5.7% last quarter), Splunk was -20.9% q/q (vs. +10.1% last quarter), and Dynatrace was -1.3% q/q (vs. +15.7% last quarter). On a y/y basis, Elastic was -21.1% y/y (vs. +22.3% last quarter), Splunk was +12.6% y/y (vs. +54.1% last quarter), and Dynatrace was +56.7% y/y (vs. +62.8% last quarter).

FIGURE 17. Elastic, Splunk, and Dynatrace Cloud - Total Visits



Source: Barclays Research, Similarweb

FIGURE 18. Elastic, Splunk, and Dynatrace Cloud - Monthly Unique Visits



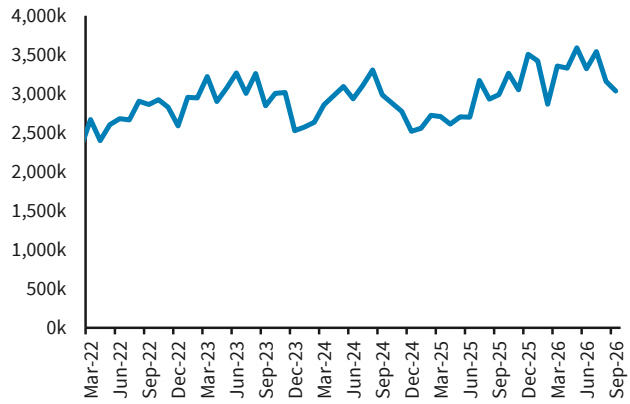
Source: Barclays Research, Similarweb

Datadog Cloud

Datadog's cloud login page has far more traffic than the other three Observability names we track and hence, we track it separately from the other vendors.

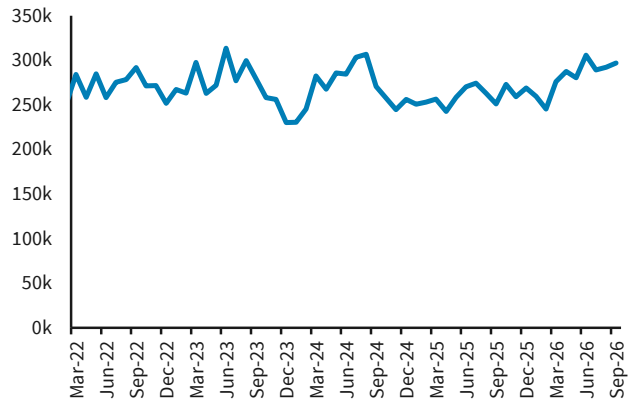
- Total visits were -4.9% q/q (vs. +6.2% last quarter) and +7.1% y/y (vs. +27.8% last quarter), while unique visits were +0.6% q/q (vs. +11.9% last quarter) and +11.4% y/y (vs. +13.2% last quarter).

FIGURE 19. Datadog Cloud Login Page - Total Visits



Source: Barclays Research, Similarweb

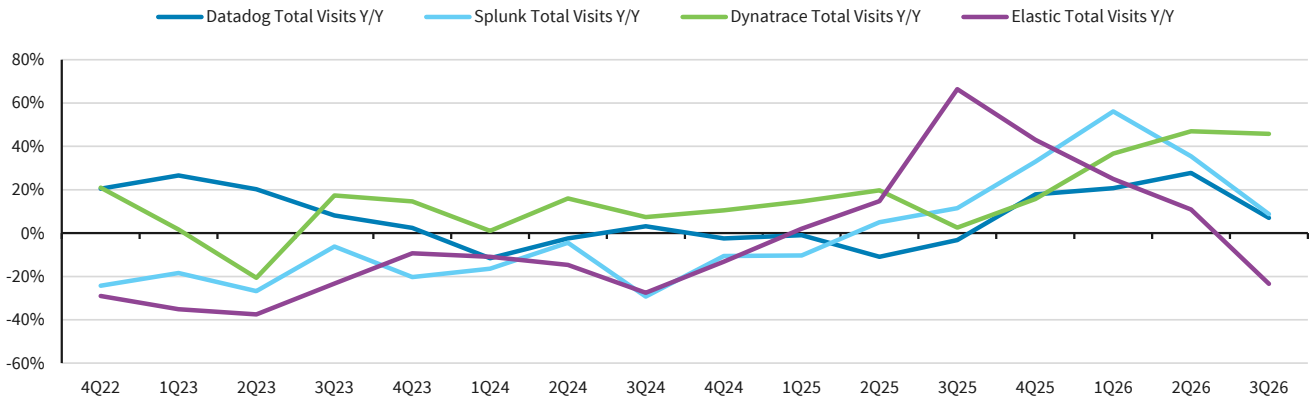
FIGURE 20. Datadog Cloud Login Page - Monthly Unique Visits



Source: Barclays Research, Similarweb

We also looked at the y/y growth changes in total visits for the Observability names. In the figure below, we can see Dynatrace (+46% vs. +47% in 2Q26) leading growth amongst these names. Elastic, however, saw an y/y decline this quarter (-23% vs. +11% in 2Q26).

FIGURE 21. Observability Total Visits Y/Y Change



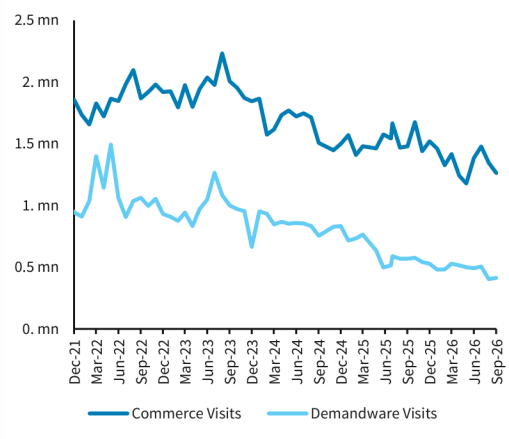
Source: Barclays Research, Similarweb

E-Commerce Platforms

We continue to track total visits to major e-commerce/omni-channel platforms, such as Shopify, Commerce, and Demandware (Salesforce). For the quarter, this group was up +10.3% q/q (vs. -1.0% last quarter) and +22.4% y/y (vs. +13.6% last quarter).

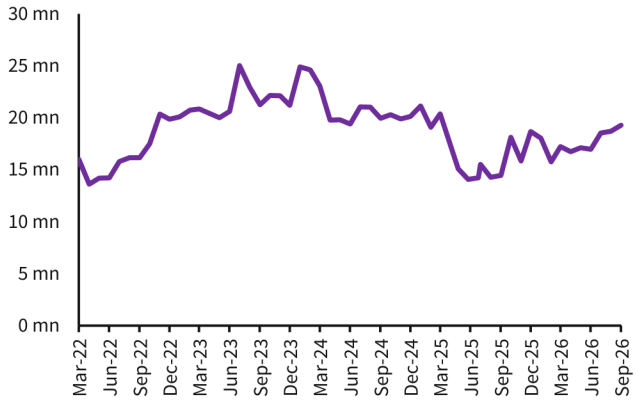
- Shopify has a much larger volume of total visits compared to peers (~10x). Total visits were up +11.2% q/q (vs. -0.3% last quarter) and up +28% y/y (vs. +17% last quarter).
- Commerce total visits were up +7.4% q/q (vs. -9.5% last quarter) and down -11.4% y/y (vs. -17.0% last quarter).
- Demandware (Salesforce) total visits were down -13% q/q (vs. +0.9% last quarter) and -23% y/y (vs. -9% last quarter).

FIGURE 22. CMRC and Demandware - Total Visits



Source: Barclays Research, Similarweb

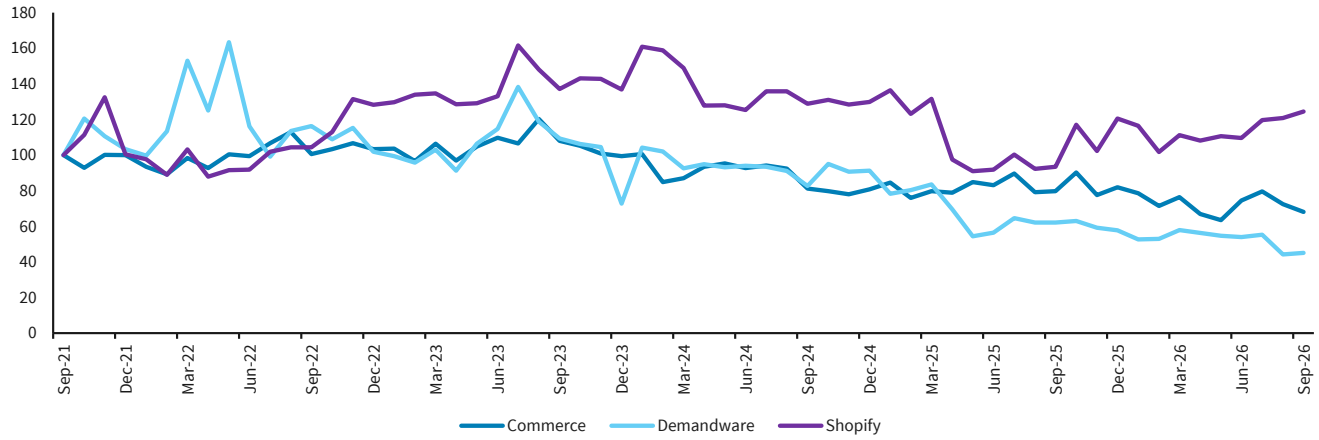
FIGURE 23. SHOP - Total Visits



Source: Barclays Research, Similarweb

Indexing the data to 100 back through September 2021, we see that Shopify has experienced the strongest growth. Demandware has seen a slight downtick in total visits, and Commerce has seen a decline this year in general but also a slight uptick in the start of Q3.

FIGURE 24. E-Commerce Platforms - Total Visits Indexed to 100



Source: Barclays Research, Similarweb

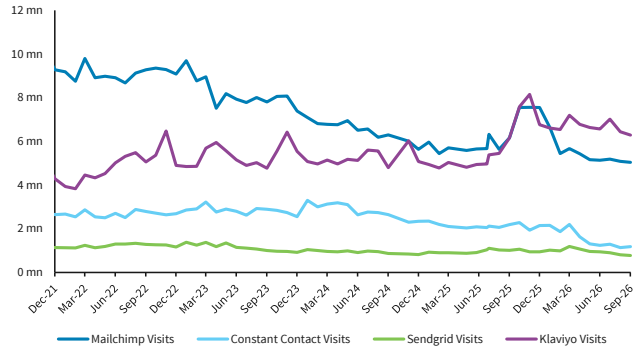
Email Marketing Traffic

We continue to track volume across email marketing vendors below.

- For total visits, Mailchimp was down -2.7% q/q (vs. -11.3% last quarter) and -15% y/y (vs. -7% last quarter), Klaviyo was down -1.2% q/q (vs. -1.8% last quarter) and +16% y/y (vs. +36% last quarter), Constant Contact (private, not covered) was down -14% q/q (vs. -33% last quarter) and -43% y/y (vs. -32% last quarter), and Sendgrid (private, not covered) was down -17% q/q (vs. -7% last quarter) and -21% y/y (vs. +5% last quarter).

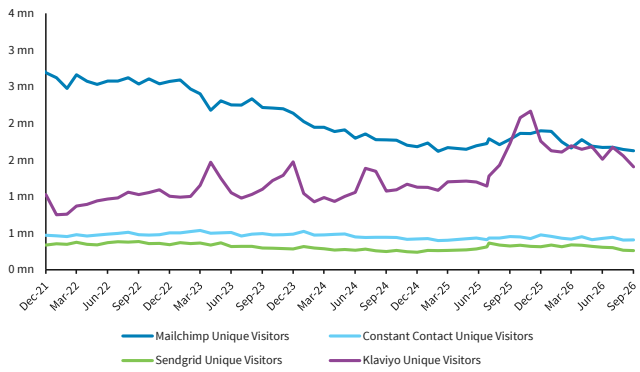
- For unique visits, Mailchimp was -3.8% q/q (vs. -3.1% last quarter) and -6.4% y/y (vs. +1.4% last quarter), Klaviyo was -4.1% q/q (vs. -1.8% last quarter) and +4.6% y/y (vs. +36.1% last quarter), Constant Contact was -2.5% q/q (vs. -1.4% last quarter) and -4.9% y/y (vs. +1.6% last quarter), and Sendgrid was -13.4% q/q (vs. -3.1% last quarter) and -19.0% y/y (vs. +11.2% last quarter).

FIGURE 25. Email Marketing - Total Visits



Source: Barclays Research, Similarweb

FIGURE 26. Email Marketing - Unique Visits



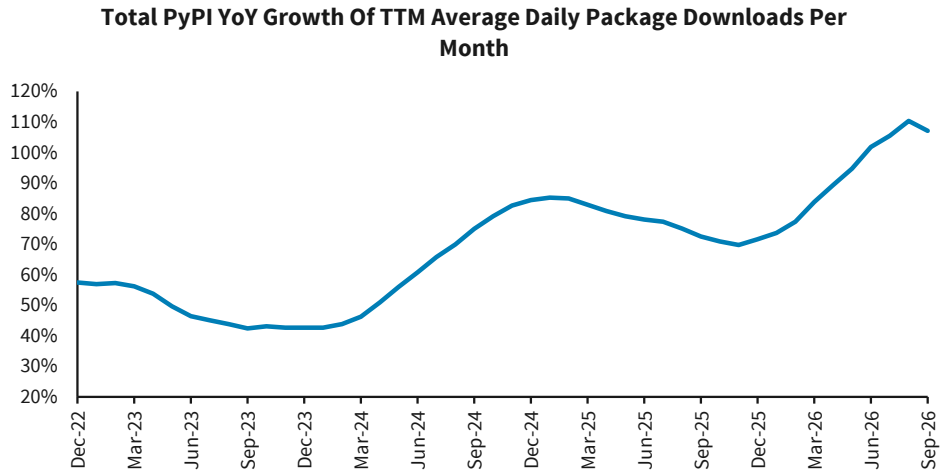
Source: Barclays Research, Similarweb

PyPI Data Tracker

Total PyPI package continues a strong 2026, although slightly softer September (which we believe is tied to the Labor Day Holiday landing in September). Total PyPI package downloads indicate a continued healthy activity throughout 2026, and usage has largely continuing to improve off softer growth in 2025 for the vendors in our analysis. We believe improving total package downloads on PyPI in general supports a more positive backdrop for DevOps and usage-based software vendors.

- **OpenAI (private, not covered) TTM PyPI downloads growth remains well above total PyPI download growth and continued strong growth over the past 6 months.** OpenAI API downloads grew on average ~272% YoY ending in September 2026 (on a TTM YoY basis). Following a healthy deceleration post very strong initial adoption trends, OpenAI TTM downloads growth have continued a healthy trend throughout 2026.

FIGURE 27. Total PyPI package downloads growth has been accelerating steadily over the past several months



Source: Barclays Research, PyPI, Google BigQuery

FIGURE 28. Amazon's (covered by Ross Sandler) acceleration in our list of vendors below appears to be the most dramatic over the past 6 months.

TTM Average Daily Package	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26	May-26	Jun-26	Jul-26	Aug-26	Sep-26
Total PyPI Growth	**75%**	**72%**	**71%**	**70%**	**72%**	**74%**	**77%**	**84%**	**89%**	**95%**	**102%**	**105%**	**110%**	**107%**
Reaccelerating Growth														
openai (private)	206%	192%	184%	179%	178%	179%	186%	197%	213%	231%	254%	268%	279%	270%
MSFT (azure-core)	26%	27%	27%	29%	31%	34%	38%	43%	46%	50%	55%	62%	68%	65%
HUBS (hubspot-api-client)	51%	51%	53%	58%	64%	69%	74%	79%	83%	86%	89%	96%	100%	95%
AMZN (boto3)	26%	19%	13%	7%	4%	2%	1%	**1%**	**7%**	**17%**	**28%**	**43%**	**57%**	**65%**
Stable Growth														
DDOG (datadog)	42%	37%	35%	35%	36%	40%	42%	45%	47%	48%	48%	49%	49%	44%
GTLB (python-gitlab)	42%	41%	41%	46%	48%	49%	46%	47%	48%	50%	50%	50%	50%	44%
GOOGL (google-cloud-core)	61%	62%	61%	61%	62%	61%	59%	57%	57%	57%	59%	62%	64%	61%
TWLO (twilio)	70%	77%	83%	90%	94%	98%	105%	118%	133%	141%	145%	149%	150%	143%
Decelerating Growth														
SNOW (snowflake-connector-python)	98%	102%	110%	112%	113%	113%	110%	110%	101%	82%	75%	76%	77%	57%
ORCL (cx-oracle plus oracledb)	115%	105%	98%	93%	89%	86%	87%	89%	86%	79%	77%	75%	71%	64%
TEAM (jira)	60%	59%	57%	55%	56%	57%	58%	62%	65%	67%	70%	70%	67%	58%
MDB (pymongo)	68%	71%	73%	71%	71%	73%	77%	81%	84%	88%	93%	96%	95%	85%
ESTC (elasticsearch)	63%	56%	45%	39%	36%	35%	36%	38%	39%	40%	40%	41%	39%	34%

The parentheses next to the company tickers are the representative PyPI package.
Source: Barclays Research, PyPI, Google BigQuery

How we utilize this data: Python is one of the fastest-growing programming languages, and as a result, PyPI downloads have grown at accelerated rates over the last few years. Therefore, along with seasonal and weekly fluctuations in the data (and with differences in how developers engage with certain software vendors and their SDKs), we think PyPI download results are best viewed on a TTM basis and are less applicable when comparing different vendor growth rates.

- **Enterprise vs. individual users offer explanation to variance between hyperscalers:** We believe that AWS package downloads are higher than Azure and Google due to the fact that AWS has the most individual / non-enterprise users amongst the three (see more on the differences between the three in Ross Sandler's note Amazon.com, Inc.: A Couple New Findings On AWS; 1/9/24).

What is PyPI?

PyPI (Python Package Index, link) is a repository of software for the Python programming language. Users of PyPI can find and install software ('packages') developed by and shared by the Python community, which saves time by reusing existing code. A 'package' is a way to organize related modules, functions, and classes hierarchically; essentially a method to bundle code. Developers can find packages on PyPI to use in their own projects, or upload their own.

- PyPI is powered by the Warehouse project, which is an open source project developed under the Python Packaging Authority (PyPA). The PyPA is an independent group of developers whose goal is to improve and maintain many of the core projects related to Python packaging.

- Developers may download packages on a daily basis because of updates to packages for new features, bug fixes, etc. Package versions typically follow 'semver' formatting (e.g., AWS boto3 is on version 1.37.25, where the last decimal represents patches for bug fixes and can be updated daily).

- For more on how we gather this data, see our note: Data Science Methods: How to get Python package (PyPI) downloads data; 2/4/25.

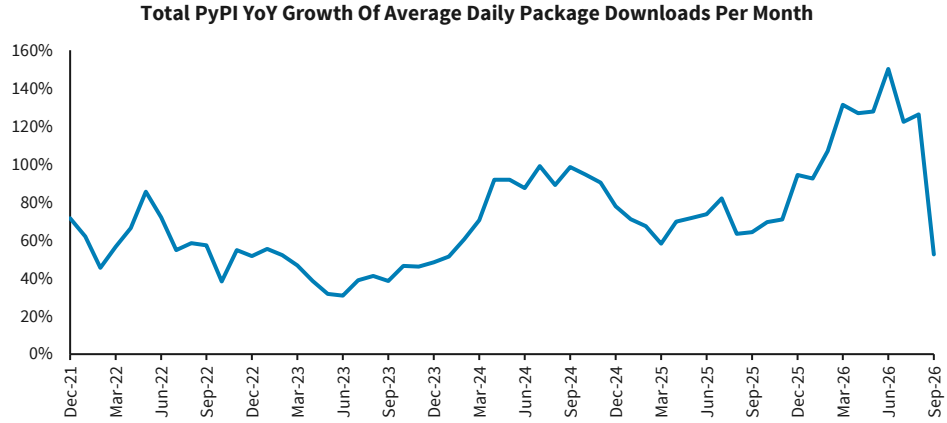
Total PyPI Package Downloads

- **Takeaway:** TTM growth continued its healthy trends in 3Q26 (see Figs above and blow), following 2025 that saw a steady deceleration. We note that in-month download growth has generally showed accelerating trends (Fig. 29 below), although there was a deceleration in September, which we believe could be tied to the 2026 Labor Day holiday falling on September 6th (compared to September 1st in 2025). Download data around holidays tracks meaningfully below normal times in our analysis.

- **What are we tracking?** PyPI offers package statistics through Google BigQuery and these downloads can be publicly accessed through the Linehaul project by streaming download logs from PyPI to BigQuery (providing download statistics of packages hosted on PyPI; for more on this check this link). The Barclays Data Science team accesses the dataset via the BigQuery API. For the individual packages we are tracking, Barclays Data Science first filters the raw data by the project name, then aggregates the downloads count for each day/month. See more in Appendix on the Barclays Data Science methodology.

- **Package description:** The data below reflects all PyPI package downloads.

FIGURE 29. Total PyPI package download growth accelerated over the last ten months, following a couple months of steady growth in 1H25.



Source: Barclays Research, PyPI, Google BigQuery

The investment views in this report are those of fundamental equity analyst Raimo Lenschow, CFA. The Data Science & Applied AI team provided supplementary data.

Appendix - PyPI Package Downloads Data

Fan Gao, CFA[v] BCI, US

This section is a Special Report that is not an equity or a debt research report under U.S. FINRA Rules 2241-2242.

PyPI offers package statistic through Google BigQuery. The tables and its pertaining data are licensed under the Creative Commons License. The "bigquery-public-data.pypi.file_downloads" table is populated through the Linehaul project by streaming download logs from PyPI to BigQuery, providing download statistics of packages hosted on PyPI. For more information, check out the guidance.

We access the dataset via the BigQuery API. For the packages we care about, we first filter the raw data by the project name, then aggregate the downloads count for each day/month. The first date we queried start from 2019-01-01.

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